Dell EMC Unisphere for PowerMax

Installation Guide

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Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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About this content

As part of an effort to improve its product lines, Dell EMC periodically releases revisions of its software and hardware. Therefore, some functions that are described in this document might not be supported by all versions of the software or hardware currently in use. The product release notes provide the most up-to-date information on product features.

Contact your Dell EMC technical support professional if a product does not function properly or does not function as described in this document.

This document was accurate at publication time. New versions of this document might be released on the Dell EMC Online Support website. Check the Dell EMC Online Support site https://support.EMC.com to ensure that you are using the latest version of this document.

Purpose

This document describes how to configure and use Unisphere for PowerMax.

Related documentation

The following publications provide additional information:

- *Dell EMC Unisphere for PowerMax Release Notes*
- *Dell EMC Unisphere for PowerMax Online Help*
- *Dell EMC Unisphere 360 Online Help*
- *Dell EMC Solutions Enabler Installation Guide*
- *Dell EMC Solutions Enabler Release Notes*
- *Dell EMC Solutions Enabler SRM CLI Product Guide*
- *Dell EMC Solutions Enabler CLI Command Reference*
- *Dell EMC PowerMax Family Security Configuration Guide*

Dell EMC technical support, documentation, release notes, software updates, or information about Dell EMC products can be obtained at https://support.emc.com (registration required) or https://www.dellemc.com/en-us/documentation/vmax-all-flash-family.htm.

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Dell EMC support, product, and licensing information can be obtained as follows:

Product information

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Technical support

Dell EMC offers various support options:

- Support by Product—Dell EMC offers consolidated, product-specific information about the Web at: https://support.EMC.com/products

  The Support by Product web pages offer quick links to the following items:
  - Documentation
  - White Papers
  - Advisories (such as frequently used Knowledgebase articles)
  - Downloads

  There is dynamic content, such as presentations, discussion, relevant Customer Support Forum entries, and a link to Dell EMC Live Chat.
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For help with any errors applying license files through Solutions Enabler, contact the Dell EMC Customer Support Center.

If you are missing a LAC letter, or require further instructions on activating your licenses through the Online Support site, contact the Dell EMC worldwide Licensing team at licensing@emc.com or call:

- North America, Latin America, APJK, Australia, New Zealand: SVC4EMC (800-782-4362) and follow the voice prompts.
- EMEA: +353 (0) 21 4879862 and follow the voice prompts.

Your comments

Your suggestions help improve the accuracy, organization, and overall quality of the documentation. Send your comments and feedback to: VMAXContentFeedback@emc.com

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Pre-installation considerations

Topics:

- Before you begin
- Environment and system requirements

Before you begin

The following sections contain steps to take and information to consider before beginning the installation:

Unisphere Initial Setup User

While installing the software, you are prompted to specify a Unisphere Initial Setup User. You can either specify the user ID you currently use to access the storage system (if User Authorization is enabled), or you can use the default user ID `smc`. If you plan on using the default ID, there must be no user IDs listed in the User Authorization database. If there are user IDs in the database (regardless of whether user authorization is enabled), then you must use SYMCLI to add the default user ID (`smc`) to the database.

For more information about users and roles, see the *Dell EMC PowerMax Family Security Configuration Guide*.

The role of the Unisphere Initial Setup User is to install and set up the Unisphere environment (create users and add roles). It is intended to be a temporary role as it enables you only to perform administrative tasks on storage systems that do not have defined user roles (authorization rules). After an Administrator or a SecurityAdmin is assigned to a storage system, the Unisphere Initial Setup User can no longer access or even see the system from the Unisphere console. Dell EMC recommends that users do not operate in this role for any longer than necessary.

Local, remote, and embedded installation options

Unisphere for PowerMax can be installed in local, remote, or embedded configurations.

In a local configuration, install the Unisphere for PowerMax software on a server that is running Solutions Enabler, which is attached to storage systems: See the Unisphere server in the following image:

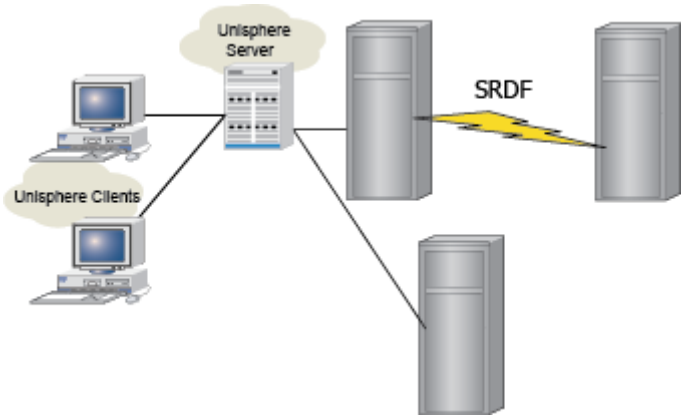


Figure 1. Local installation: Unisphere for PowerMax

In a remote configuration, install the Unisphere software on a server that is connected to the SYMAPI server. See the Unisphere server in the following image:

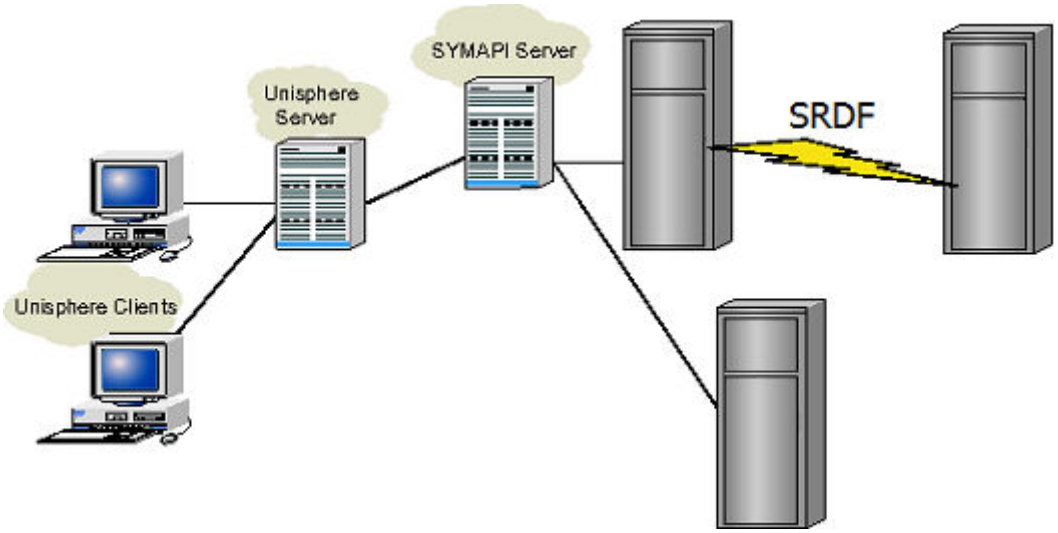


Figure 2. Remote installation: Unisphere for PowerMax on a server connected to a remote SYMAPI server

Alternatively, starting with HYPERMAX OS Q3 2015, you can manage storage systems running HYPERMAX OS 5977 or later using the Embedded Management (eManagement) container application. eManagement embeds management software (Solutions Enabler, SMI-S, Unisphere for PowerMax) on the array, enabling management of the array without requiring software installed on a host.

Unisphere for PowerMax virtual appliance

The Unisphere for PowerMax virtual appliance (vApp) is available for ESX/ESXi V5.0 (and higher) servers in a VMware environment.

For instructions on installing the appliance, see the *Dell EMC Virtual Appliance Manager Installation Guide*.

Authentication

Unisphere for PowerMax supports the following types of authentication:

Windows (local and domain-based)	Users log in by specifying the Windows domain, username, and password (if they have a Windows account on the SMAS server).
LDAP	Users log in with their LDAP-SSL username and password (if they have a user account that is stored on an LDAP-SSL server). To use this method, a Unisphere Administrator or SecurityAdmin sets up LDAP-SSL authentication in Unisphere for PowerMax. The *Unisphere for PowerMax Online Help* contains instructions on performing these tasks.
Local Unisphere users	Users log in with their Unisphere username and password (if they have a local Unisphere user account). To use this method, a Unisphere Initial Setup User, Administrator, or SecurityAdmin creates a local Unisphere user account for the user. Local user accounts are stored locally on the SMAS server host and work in a similar way as the other methods used to validate user credentials. The *Unisphere for PowerMax Online Help* contains instructions on performing these tasks.
X.509 certificate-based user authentication	Certificate-based user authentication using X.509 certificates is supported on fresh installations on Windows and Linux servers. You can use a certificate that is issued by a trusted public third-party certificate authority (CA) to authenticate your identity when using the Unisphere for PowerMax web client or REST API interfaces. The use of digital identity smartcards such as Common Access Card (CAC) and Personal Identity Verification (PIV) as part of a multifactor authentication process is also supported. You can enable certificate-based user authentication as part of the installation and after you have confirmed your intentions, the choice becomes irreversible. Import the CA certificates before certificate-based user authentication can be used.

The *Dell EMC PowerMax Family Security Configuration Guide* provides more details about authentication, authorization, and other issues related to security.

Environment and system requirements

The Unisphere web server is supported on the platforms that are listed in Server requirements and limitations. Unisphere for PowerMax can be used in remote client/server mode to communicate with SYMAPI servers on other systems, including supported versions of AIX, Linux, Solaris, HP-UX, z/OS, and Windows.

Network requirements

Before installing Unisphere for PowerMax, verify that the host is in a properly configured IP network and that it can resolve IP addresses and hostnames.

When you are installing Unisphere for PowerMax, the use of IPv6 addresses is supported.

Solutions Enabler

Solutions Enabler 64-bit V9.2.1 is required in local and client/server modes. In client/server mode, the `storsrvd` daemon must be running on the server.

Supported PowerMaxOS/HYPERMAX OS/Enginuity versions

Unisphere for PowerMax V9.2.1 supports:

- Storage systems running PowerMaxOS 5978 (Q1 2021 release, or earlier)
- Storage systems running HYPERMAX OS 5977
- Storage systems running Enginuity 5876

Host systems and storage operating environment

Unisphere for PowerMax runs on a wide range of operating systems and works with certain Enginuity, HYPERMAX OS, and PowerMaxOS versions. For detailed interoperability information, see E-Lab Interoperability Navigator at https://support.EMC.com.

Concurrent users

Unisphere for PowerMax supports up to 10 concurrent users, with some performance degradation occurring with more than 5 storage admin users.

Note the following recommendations on REST API client limits, which must be adhered to in order to achieve optimal performance:

- A maximum of 5 REST API clients can connect to the Unisphere server concurrently.
- A maximum of 50 GET calls can be made to the Unisphere server per second. These can be made up of performance REST API GET and POST calls and non-performance REST API GET calls.
- A maximum of 5 non-performance POST or PUT calls can be run against the Unisphere server concurrently.
- It is recommended that you use bulk REST API calls, where possible.

NOTE: If the recommendation above are not followed, it might result in performance degradation of the Unisphere for PowerMax server.

VMware and Hyper-V guests

Unisphere for PowerMax is supported on a guest operating system on the following platforms, provided the guest is listed in the EMC Support Matrix and in VMware/Hyper-V support documentation, and Unisphere for PowerMax supports the platform:

- VMware ESX/ESXi Server versions 5.0, 5.1, 5.5, 6.0 and 6.5 (and all updates of each version)
- Windows Server 2016
- Virtual machines must not run on shared memory or resources.

NOTE: The guest must provide the same CPU, memory, disk, gatekeeper, and other requirements as if Unisphere for PowerMax were installed on a physical machine.

Database Storage Analyzer requirements

Database Storage Analyzer (DSA) monitors the performance of the following databases:

- Oracle 12 (not container), 12 (both CDB and PDB), 18c, and 19c
  NOTE: DSA supports Oracle running on Linux operating systems only.

  For CDB DSA guest user name should be started with c## and 12 RAC.
- SQL Server 2012, 2014, 2016, 2017, and 2019
  NOTE: DSA supports SQL Server running on Windows operating systems only.

The following operating systems are supported:

- AIX, version 5.2 and above (64-bit)
- Sun Solaris 10 and above (64-bit, SPARC only)
- Red Hat Enterprise Linux 5 and above (64-bit)
- Linux SuSE 10 and higher
- Oracle Linux Server release 5 and higher

The following storage platforms are supported:

- Storage systems running Enginuity 5876
- Storage systems running HYPERMAX OS 5977
- Storage systems running PowerMaxOS 5978

You can use VMware VMDK and RDM virtual environments. Other virtual environments (Solaris containers, AIX VIO, VPLEX, and so on) are not supported.

10 GB is required on the monitored database host for storing the mapping output.

A monitored Oracle database must have an Oracle diagnostic pack license.

To use Database Storage Analyzer that is packaged with Unisphere for PowerMax to monitor a SQL Server database, you must install the latest Database Storage Analyzer Listener on the monitored environment. The procedure to install the Database Storage Analyzer Listener is described here: Upgrading Database Storage Analyzer Listener.

EMC ControlCenter

Unisphere cannot be installed on the same host as EMC ControlCenter agents. This restriction is because Unisphere requires a 64-bit version of Solutions Enabler and ControlCenter agents require a 32-bit version of Solutions Enabler.

Client screen requirements

The minimum client screen resolution that is required is 1280 x 1024.

Client browser requirements

The Unisphere for PowerMax client is viewed in a browser and does not use dynamic ports, so it works with most VPN solutions.

NOTE: The preferred browsers for Unisphere for PowerMax 9.2.1 are Chrome and Firefox.

The following table defines the client browser requirements:

Table 1. Unisphere for PowerMax client browser requirements

Browser	Version	Memory [a]	Platform
Chrome[b]	76.0.3809.100	600 MB	Windows
Firefox	68.0		
Internet Explorer	11.0 (Desktop only)		
Microsoft Edge	44		

a. Required for the browser as it runs on the machine from which the console is launched .
b. Not supported in embedded NAS (eNAS) environments.

Server requirements

The table defines the client requirements for Unisphere for PowerMax.

Table 2. Server requirements and limitations

Operating system[a]	Maximum volume count	Minimum processor specification	Minimum available memory[b]	Minimum available disk space	Maximum managed storage system count
Windows Linux	0–64,000	Minimum of a two-core, 2.5 GHz processor	12 GB	120 GB	10
Windows Linux	64,001–128,000	Minimum of a two-core, 2.5 GHz processor	16 GB	140 GB	15
Windows Linux	128,001–256,000	Minimum of a six-core, 2.5 GHz processor	24 GB	180 GB	20

a. For specific operating system information, see E-Lab™ Interoperability Navigator at: https://support.EMC.com
b. The minimum amount of memory that is required on the server (before installation) to successfully run Unisphere for PowerMax. It is not the total amount of memory on the server. If a server has 12 GB of physical memory that is installed but there is only 6 GB available before Unisphere for PowerMax is installed, you may experience performance issues with the application. If running on a Virtual Machine, this amount must be dedicated memory, not shared memory.

The following packages are prerequisites for Unisphere for PowerMax:

- Microsoft Visual C++ 2010 x64 Redistributable
- Microsoft Visual C++ 2013 Redistributable (x64)
- Microsoft Visual C++ 2015 Redistributable (x64)

If they are not already installed, then they are installed as part of the Unisphere for PowerMax installation.

Other requirements

- DNS must be configured correctly on the host.
- The host system must have connectivity to one or more storage systems to add users and set permissions in Unisphere.
- Unisphere for PowerMax can be installed on the same system as the SYMAPI server, or on a system connected to the SYMAPI server. For more information, see Local installation: Unisphere for PowerMax and Remote installation: Unisphere for PowerMax on a server connected to a remote SYMAPI server, respectively.
- Before you attempt to install Unisphere for PowerMax on Linux, verify that the following packages are installed:

  Red Hat Enterprise Linux 6.0, or later:

  - `libstdc++-devel-4.4.4-13.el6.x86_64.rpm`
  - `libstdc++-4.4.4-13.el6.x86_64.rpm`
  - `libaio-0.3.107-10.el6.x86_64.rpm`
  - `ncurses-libs-5.7-3.20090208.el6.x86_64.rpm`

SUSE Linux Enterprise Server 15, or later:

- `insserv-compat-0.1-4.3.1.noarch.rpm`

Not all installer packages include these libraries by default, so verify their presence before running the installation program. Do not attempt to run the installation process without them. Without these libraries, the product might fail to install or not function properly.

- To install PostgreSQL on Linux, a minimum of 2 GB of shared memory must be available.

  For more information about how to configure the size of a shared memory segment, using the `shmmax` kernel parameter, see the documentation for your operating system.

- If you are changing your SYMAPI connection type, such as changing from a local connection to a client/server configuration, you must perform a fresh installation.
- Before installing Unisphere for PowerMax on a host running Linux, ensure that the maximum number of open file descriptors that are allowed is at least 4096.
- For installations on Microsoft Windows, Unisphere must be installed on a volume which has Windows short file names (8dot3) enabled.

  To determine if short names are enabled for a volume, use:

  ```
  fsutil 8dot3name query Volume_Drive_Letter
  ```

  Return values are:

  Volume state = 0 (8dot3 is enabled)

  Volume state = 1 (disabled)

  To enable 8dot3, use:

  ```
  fsutil 8dot3name set Volume_Drive_Letter 0
  ```

  For example, to enable 8dot3 on the C drive

  ```
  fsutil 8dot3name set C: 0
  ```

- Ensure that your system is configured to use a supported operating system, language, and keyboard.
- Before installing Unisphere on a Microsoft Windows system, ensure that the language for the system location is correct to avoid indecipherable characters on the install screens. To check the language, go to **Control Panel** > **Region and Language** > **Administrative tab** > **Language for non-Unicode programs**.
- Unisphere for PowerMax supports Active Directory over LDAP and Open LDAP version 2.4 and later.

2

Installing Unisphere for PowerMax

Topics:

- Preinstallation configuration
- Installing Unisphere for PowerMax
- Completing the installation
- Launching Unisphere for PowerMax

Preinstallation configuration

Before you install Unisphere for PowerMax, complete the following tasks if they are appropriate for your installation environment:

- Setting the event and statistics daemons to autostart on Linux
- Setting up the PostgreSQL user on Linux
- Configuring networking on Linux

Setting the event and statistics daemons to autostart on Linux

To install Unisphere for PowerMax on a local Linux 64-bit host, complete this task.

To set the event and statistics daemons to autostart:

1. Set the event and statistics daemons to autostart using the following commands:

```
stordaemon install storevntd -autostart
stordaemon install storstpd -autostart
```

2. Start the same daemons using the following commands:

```
stordaemon start storstpd
stordaemon start storevntd
```

Setting up the PostgreSQL user on Linux

Before starting a new installation of Unisphere for PowerMax or Unisphere 360 on Linux, the PostgreSQL user (`postgres`) and group (`postgres`), must be present. Also, the user must be a member of the `postgres` group.

Use the following commands to create the `postgres` group and add the `postgres` user to the `postgres` group:

```
groupadd postgres
useradd -g postgres postgres
```

Configuring networking on Linux

In the `/etc/hosts` file, comment out the following text:

```
::1
```

This change is needed for both IPv4 and IPv6 configurations.

Configuring file system mountpoints on Linux systems

About this task

When mounting the `/tmp` directory, you must remove the `noexec` parameter.

If you do not want to remove the `noexec` parameter for the `/tmp` directory, set an alternative directory for the installer.

For `sh`, `ksh`, `bash`, and `zsh`:

```
$ IATEMPDIR=new_temp_dir
$ export IATEMPDIR
```

For `csh` and `tcsh`:

```
$ setenv IATEMPDIR new_temp_dir
```

Installing Unisphere for PowerMax

Prerequisites

NOTE: Before installing Unisphere for PowerMax 9.2.1, you must install Solutions Enabler 9.2.1 and run the `symcfg discover` command.

About this task

To install Unisphere for PowerMax:

Steps

1. Accessing the software: Step 1: Accessing the software .
2. Installing the software: Step 2: Installing the software.
3. Installing Unisphere for PowerMax using the wizard: Step 2A: Installing Unisphere for PowerMax using the wizard.
4. Installing Unisphere for PowerMax from the command line: Step 2B: Installing Unisphere for PowerMax from the command line.
5. Installing Unisphere for PowerMax using silent install: Step 2C: Installing Unisphere for PowerMax using silent install

Step 1: Accessing the software

About this task

The Unisphere for PowerMax software and documentation is available on the Dell EMC support website at: https://support.emc.com/

It consists of the following:

- For Windows 64-bit:

  ```
  UNIVMAX_Vx.x.x.y_WINDOWS_X86_64.exe
  ```

- For Linux 64-bit:

  ```
  UNIVMAX_Vx.x.x.y_LINUX_X86_64.bin
  ```

  In the file names above, the x represents the release number and the y represents the software build number. The build number varies based on when the software was built.

- *Unisphere for PowerMax Online Help*
- *Unisphere for PowerMax Release Notes*

- This installation guide

To access the software from online support:

Steps

1. On the Dell EMC support website:
   a. Search for **`Unisphere for PowerMax`**.
   b. Locate the appropriate kit and download it to a directory on the host drive.
2. Continue with Step 2: Installing the software.

Step 2: Installing the software

About this task

You can install Unisphere for PowerMax using an installation wizard, as described in Step 2A: Installing Unisphere for PowerMax using the wizard, or from the command line, as described in Step 2B: Installing Unisphere for PowerMax from the command line.

Step 2A: Installing Unisphere for PowerMax using the wizard

About this task

To install Unisphere for PowerMax using the wizard:

Steps

1. Save all files and exit all applications.
2. Change the directory to the location of the kit and run the appropriate executable file for your operating system.
   To install Unisphere for PowerMax using the wizard in Linux, enable the display manager and use the `-i swing` switch, for example:

   ```
   ./UNIVMAX_Vx.x.x.x_LINUX_X86_64.bin -i swing
   ```

   If the display manager is not enabled or the `-i swing` switch is not used, the installer defaults to console mode.

   When using the non-root user feature on a Linux installation, the Unisphere installer picks up the user name running the SE daemons and assigns that to the Unisphere Java process.
3. On the **Introduction** page, review the recommendations and click **Next**.
4. On the **Choose Install Folder** page, click **Next** to accept the default directory, or click **Choose** to select another directory.

   NOTE: On Red Hat Enterprise Linux 8.0 and above, installing on a non-root partition is not supported.
5. On the **Initial Setup User and SymAPI Configuration** page:
   a. Specify a **Unisphere Initial Setup User** name (default is `smc`). You can either accept the default or use your Windows username if you have enabled authorization on the Unisphere server host. For more information about the role of the Unisphere Initial Setup User, see Unisphere Initial Setup User.
   b. Specify a **SymAPI Connection Type** for the initial login based on the following:
      - **Local**—On a host that has Solutions Enabler installed and is attached to PowerMax systems.
      - **Remote**—On a remote server that is connected to a SYMAPI server, specify the SYMAPI server **Node Name** and **Net Port**. The default net port for SYMAPI is 2707.

      For more information about each of the connection types, see Local, remote, and embedded installation options.
   c. Click **Next**.
6. On the **Ports Configuration** page, specify the ports to use or enable when connecting to the Unisphere server.
   The default ports of the Unisphere server (**HTTPS Port No**) and performance database (**DB Port No**) are shown. To use the default ports, verify that they are available, and click **Next** to accept and enable them. Or you can specify and enable different ports, and click **Next**.
7. Optional: On the **X.509 Certificate-based Client Authentication** page, configure certificate-based user authentication options. The default behavior is for X.509 certificate-based authentication to remain disabled.

To enable X.509 certificate-based authentication, complete the following steps:

a. Select **Enable certificate based client authentication?**.
b. Specify whether to use the CN or UPN as the source of user identity.

   NOTE: The following special characters are stripped from the authorization rule: `@:?;|<>[]+=,*/\`

c. To specify Unisphere 360 X.509 common name, select **Define Unisphere 360 X.509 Common Name**.
d. In the **Unisphere 360 X.509 Common Name** field, type the Unisphere 360 X.509 common name.
e. Click **Next**.
f. On the **Confirm Client Authentication** dialog box, click **OK**.

If you enable X.509 certificate-based authentication, after installation completes, import the certificate and start SMAS manually. For more information, see Importing CA certificates.

8. To install Unisphere for PowerMax on a host with more than one network interface configured, you are prompted to choose the IP address to which Unisphere for PowerMax binds. On the **Listening (bind) Address Selection** page, select the server protocol (IPv4 or IPv6), select the IP address, and then click **Next**.
9. If the disk space and/or memory requirements are not met, a warning is displayed. If you want to continue with the installation regardless of the consequences described in the message, click **OK**.
10. On the **Pre-Installation Summary** page, verify the summary information, and click **Install** to continue the installation.
11. On the **Install Complete** page, click **Done**.
12. Refer to Completing the installation for instructions on completing the installation.

Step 2B: Installing Unisphere for PowerMax from the command line

About this task

Online help is available when installing from the command line by entering `-?`.

To install Unisphere for PowerMax from the command line:

Steps

1. Save all files and exit all applications.
2. Change the directory to the location of the kit.
3. Run the following command to install Unisphere for PowerMax on Windows:

```
kit_name -i console
```

Run the following commands to install Unisphere for PowerMax on Linux:

```
chmod +x kit_name
./kit_name -i console
```

Where *kit_name* is one of the following operating system-specific kit names:

- For Windows 64-bit:

```
start /wait UNIVMAX_Vx.x.x.y_WINDOWS_X86_64.exe
```

  The start /wait option starts the installer and waits for it to end.

- For Linux 64-bit:

```
UNIVMAX_Vx.x.x.y_LINUX_X86_64.bin
```

In the file names above, the x represents the release number and the y represents the software build number. The build number varies based on when the software was built.

When using the non-root user feature on a Linux installation, the Unisphere installer picks up the user name running the SE daemons and assigns that to the Unisphere Java process.

4. On the **Introduction** panel, review the recommendations and press **Enter**.
5. On the **Choose Install Folder** panel, press **Enter** to accept the default directory, or specify a different directory and press **Enter**.

NOTE: On Red Hat Enterprise Linux 8.0 and above, installing on a non-root partition is not supported.

6. On the **postgres User and Group** panel, you are instructed to create the `postgres` group and `postgres` user before continuing. When complete, press **Enter**.
7. On the **Initial Setup User and SymAPI Configuration** panel, complete the following steps:
   a. Specify a **Unisphere Initial Setup User name** (default is `smc`).

      You can either accept the default username or if you have enabled authorization on the Unisphere server host, use your Windows username. For more information about the role of the Unisphere Initial Setup User, see Unisphere Initial Setup User.
   b. Specify a **SymAPI Connection Type** for the initial login based on the following:
      - **Local**—On a host with Solutions Enabler installed and attached to PowerMax systems.
      - **Remote**—On a remote server that is connected to a SYMAPI server, specify the SYMAPI server **Node Name** and **Net Port**. The default net port for SYMAPI is 2707.

      For more information about each of the connection types, see Local, remote, and embedded installation options.
8. On the **Ports Configuration** panel, specify the ports to use or enable when connecting to the Unisphere server. The default ports of the Unisphere server (**HTTPS Port No**) and performance database (**DB Port No**) are shown. To use the default ports, verify that they are available, and click **Next** to accept and enable them. Or you can specify and enable different ports, and click **Next**.
9. On the **Unisphere Lockbox Password** panel, accept the default settings.
10. Optional: On the **X.509 Certificate-based Client Authentication** panel, configure certificate-based user authentication options. The default behavior is for X.509 certificate-based authentication to remain disabled.

    To enable X.509 certificate-based authentication, complete the following steps:
    a. At the **Enable certificate based client authentication?** prompt, type **2** for Yes.
    b. Specify whether to use the CN or UPN of the client certificate as the certificate alias.

       NOTE: Instances of the following special characters are stripped from the alias:

       `@:?;|<>[]+=,*/\`
    c. Press **Enter** to continue.
    d. Optional: To specify Unisphere 360 X.509 common name, type **2** and type the Unisphere X.509 common name. Press **Enter** to continue.
    e. On the **Confirm Client Authentication** panel, confirm your selection.

    If you enable X.509 certificate-based authentication, after installation completes, import the certificate and start SMAS manually. For more information, see Importing CA certificates.
11. On the **Listening(bind) Address Selection** panel, to install Unisphere for PowerMax on a host with more than one network interface configured, you are prompted to choose the IP address to which Unisphere for PowerMax will bind and specify if it is an IPv4 or an IPv6 address.
12. If the disk space and/or memory requirements are not met, a warning is displayed. If you want to continue with the installation regardless of the consequences described in the message, type **1**.
13. On the **Pre-Installation Summary** panel, verify the summary information, and the press **Enter** to continue the installation.
14. On the **Install Complete** panel, press **Enter**.
15. Refer to Completing the installation for instructions on completing the installation.

Step 2C: Installing Unisphere for PowerMax using silent install

About this task

Silent install provides a non-interactive mode of installation using a response file. This enables installation simultaneously on multiple hosts and helps to automate the install process.

Typical response file contents are in the form of key-value pairs. The keys of the response files are interpreted by the installer as InstallAnywhere variables and their values are replaced during runtime. Typical contents of a response file are as follows:

```
EMC_SUNAME=smc
EMC_HTTPSPORT= [PORT NUM]
EMC_DBPORT=[PORT NUM]
```

```
EMC_CONNTYPE=[Local | Remote]
EMC_NODENAME=[localhost| remote host]
EMC_NETPORT=[PORT NUM]
DEFAULT_PASSWORD=[true]
BIND_ADDRESS=[IP ADRESS]
IS_RESPONSE_FILE=[always true]
USER_INSTALL_DIR=[install directory]
```

The following list contains definitions of the variables:

- EMC_SUNAME—Initial Setup Username.
- EMC_HTTPSPORT—Custom HTTPS port value.
- EMC_DBPORT—Database port value.
- EMC_CONNTYPE—SYMAPI connection type. Can be either Local or Remote.
- EMC_NODENAME—Sets the IP of the host on which SYMAPI is running.
- EMC_NETPORT—Port value for Remote SYMAPI connection type.
- DEFAULT_PASSWORD—This value should always be true.
- BIND_ADDRESS—The bind address used by JBoss.
- IS_RESPONSE_FILE—Mandatory variable to inform the installer that the response file is passed in silent install mode and input should be taken from the response file.
- USER_INSTALL_DIR—Installation path.

To install Unisphere for PowerMax using silent install:

Steps

1. In the install directory, create a response file with a `.properties` extension and add the parameters listed above. The following example shows a response file with default options:

```
EMC_SUNAME=smc
EMC_HTTPSPORT=8443
EMC_DBPORT=3324
EMC_CONNTYPE=Local
EMC_NODENAME=localhost
EMC_NETPORT=2707
DEFAULT_PASSWORD=true
BIND_ADDRESS=198.51.100.99
IS_RESPONSE_FILE=TRUE
USER_INSTALL_DIR=C:\\Program Files\\EMC
```

   NOTE: In the case of Linux the install directory would be `/opt/emc`.

   NOTE: On Red Hat Enterprise Linux 8.0 and above, installing on a non-root partition is not supported.

2. Invoke a silent install from the command line by typing one of the following commands:
   - For Linux:

     ```
     ./UNIVMAX_version.bin -i silent [-f response_file]
     ```

   - For Windows:

     ```
     UNIVMAX_version.exe -i silent [-f response_file]
     ```

   NOTE: For Windows you can use `Start /wait` to end the command execution:

   ```
   start /wait UNIVMAX_version.exe -i silent -f response_file
   ```

Completing the installation

About this task

This section describes the tasks to perform after the installation has been completed:

- Configuring language and regional settings: Configuring language and regional settings
- Configuring third-party utilities: Configuring third-party utilities

- Importing CA certificates: Importing CA certificates
- Starting and stopping the performance database: Starting and stopping the performance database
- Starting and stopping the SMAS service: Starting and stopping the SMAS service

Installing Database Storage Analyzer Listener

About this task

Unisphere for PowerMax includes the Database Storage Analyzer (DSA) application that provides a database to storage performance troubleshooting solution for Oracle and SQL server databases running on PowerMax storage systems.

Install the DSA Listener on a Microsoft SQL Server host. The following Windows Server versions are supported:

- Windows Server 2012
- Windows Server 2012 R2
- Windows Server 2016
- Windows Server 2019

> **NOTE:**
>
> If using Windows 2019, you must install the OpenSSH client and server separately. Use the instructions provided by Microsoft, at https://docs.microsoft.com/en-us/windows-server/administration/openssh/openssh_install_firstuse, to install and configure OpenSSH. By default, it is installed on port 22, and this is the port that you should use when installing DSA.

You must have Administrator privileges to complete the installation.

Before you begin, ensure that no processes other processes are using the default Database Storage Analyzer Listener port (22200). If that port is in use, the installation fails.

Steps

1. Launch the installer.
2. Proceed through the installation wizard, accepting the default values. The installation wizard installs and starts the required services.

Configuring language and regional settings

About this task

For Windows users outside the United States, you must configure your regional and language settings to English (United States) if you plan on using the graph feature of the Quality of Service (QoS) and Replication Monitors.

The exact steps that you need to follow might vary from the steps that are listed here, depending on the version of Windows you are using.

To configure the language and regional settings:

Steps

1. Stop the EMC SMAS service (if it is running). See Starting and stopping the SMAS service.
2. In the Windows **Control Panel**, double-click **Regional and Language Options**.
3. On the **Regional Options** tab, set the **Standards and formats to English (United States)**.
4. On the **Advanced** tab, set the **Language for non-Unicode programs to English (United States)** and select **Apply all setting to the current user account and to the default user profile**.
5. Restart the service.

Configuring third-party utilities

About this task

To ensure that third-party software utilities (including virus scanning and backup tools) do not affect the operation of the Unisphere for PowerMax software that is installed on the host, you might need to carry out some additional configuration.

Exclude the `install_dir\SYMAPI\stp\spa` directory and all subdirectories from any virus scan.

Exclude the performance database directory, `install_dir\SMAS\jboss\standalone\data\pgsql\data`, and all subdirectories, from the actions performed by any third-party software utility.

NOTE: Not following this advice may lead to data corruption in the performance database or problems with data collection.

Importing CA certificates

About this task

If you chose to use certificate-based user authentication at installation time, import the CA certificates before you login.

You must have root/administrator permissions to complete this task.

When certificate-based user authentication is chosen at install time, the SMAS service does not start automatically after installation completes. After you have imported the CA certificates, manually start the SMAS service.

For more information about starting the SMAS service refer to Starting and stopping the SMAS service.

To import a CA certificate:

Steps

1. Open the domain.xml.init file.
   For Linux:

   ```
   install_dir/jboss/domain/configuration/domain.xml.init
   ```

   For Windows:

   ```
   install_dir\jboss\domain\configuration\domain.xml.init
   ```

2. Search for the instance of `"ssl key-alias="tomcat" password=password"`.
3. Take note of the value of the `password` attribute.
4. For each root CA and intermediate CA certificate, run the appropriate commands.
   For Windows:

   ```
   cd install_dir\SMAS\jboss\bin
   service.bat importcacert password CA_cert_alias CA_file_path
   ```

   For Linux:

   ```
   cd install_dir/SMAS/jboss/bin
   initcst.sh importcacert password CA_cert_alias CA_file_path
   ```

   Where:
   - `password` is the password of the keystore, discovered in 3
   - `CA_cert_alias` is the alias of the CA cert chosen at installation time.

     Note that when either the CN or UPN is chosen as the alias for the CA cert, instances of the following special characters are stripped from the alias: @:?;|<>[]+=,*/\
   - `CA_file_path` is the path to the CA certificate file.

Starting and stopping the performance database

About this task

If required to do so, you can start and stop the performance database manually.

Windows

About this task

To start or stop the EMC_smasdb service in Windows, use the Control Panel (**Administrative Tools** > **Services**).

Alternatively, you can use the following commands to start or stop the Windows services using the CLI:

```
net start "EMC_smasdb"
net stop "EMC_smasdb"
```

Linux

About this task

Start the performance database before starting the SMAS service. In the same way, stop the SMAS service before stopping the performance database.

To start or stop the performance database (PostgreSQL) in Linux, navigate to this directory: `install_dir/SMAS/jboss/standalone/data/pgsql` and use the following commands:

Action	Command
Start	`./smasdb_unix_helper.sh -S`
Stop	`./smasdb_unix_helper.sh -k`

Starting and stopping the SMAS service

About this task

If required to do so, you can start and stop the SMAS service manually.

Windows

About this task

To start or stop the EMC Symmetrix Management Application Server in Windows, use the **Services** panel. Click **Control Panel** > **Administrative Tools** > **Services** to open the **Services** panel.

Alternatively, you can use the following commands to start or stop the Windows services using the CLI:

```
net start "EMC Symmetrix Management Application Server"
net stop "EMC Symmetrix Management Application Server"
```

Linux

About this task

Start the performance database before starting the SMAS service. In the same way, stop the SMAS service before stopping the performance database.

To start or stop the SMAS service in Linux, change to this directory: `/etc/init.d` and use the following commands:

Action	Command
Start server	`./smas start`
Stop server	`./smas stop`

Launching Unisphere for PowerMax

About this task

NOTE: After performing any of the following operations, it can take up to 30 minutes (depending on the size of the storage environment) before Unisphere for PowerMax becomes available:

- Installing
- Upgrading
- Rebooting
- Restarting the SMAS service

To launch Unisphere for PowerMax:

Steps

1. Type one of the following URLs in a browser:

   ```
   https://host_IP:port_number
   ```

   or

   ```
   https://host_name:port_number
   ```

   The port number is configured during installation. The default HTTPS port number is 8443.

   If the host IP address is an IPv6 address, surround the IP address with square brackets, for example:

   `https://[2001:db8:ffff:ffff:ffff:ffff:ffff:ffff]:8443`

   If the host IP address is an IPv4 address, type the IP address as normal, for example:

   `https://198.51.100.255:8443`

2. Do one of the following:
   - At the login window, type the Unisphere Initial Setup User username and password, and click **Login**.

     The default username for the Unisphere Initial Setup User is `smc` and the default password is `smc`.

   - If you are using certificate-based user authentication, click **Options** to display more options, and select **Use Client Certificate**. If the Unisphere for PowerMax login window does not display correctly, ensure that the CA certificates have been imported correctly.

3. Click **Login**.

3

Upgrading Unisphere for PowerMax

Topics:

- Upgrade paths
- Before upgrading Unisphere for PowerMax
- Running the Unisphere for PowerMax upgrade
- After upgrading Unisphere for PowerMax
- Upgrading Database Storage Analyzer Listener

Upgrade paths

You can upgrade directly to Unisphere for PowerMax V9.2.x from Unisphere for PowerMax V9.1.x.

NOTE: To upgrade to V9.2.x from a version lower than V9.1.x, first upgrade to V9.0.x, and then to V9.1.x.

NOTE: Unisphere for PowerMax release 9.1.0 onwards uses PostgreSQL version 10.

On SUSE Linux Enterprise Server (SLES) platforms, PostgreSQL version 10 is supported only on SLES version 12 or later.

On Red Hat Enterprise Linux Server (RHEL) platforms, PostgreSQL version 10 is supported only on RHEL version 6, or later.

The installation program automatically upgrades your installation according to the settings used in the previous installation, including port and connection type.

The following Unisphere for PowerMax settings from your previous installation are maintained through the upgrade process:

- Users
- Alert policies and threshold settings
- Email settings: Mailing list and SMTP configuration
- Performance dashboard settings
- Queries

When you are upgrading from V8.x, LDAP/AD configuration information is retained.

Before upgrading Unisphere for PowerMax

Prerequisites

NOTE: Before upgrading to Unisphere for PowerMax 9.2.1, you need Solutions Enabler 9.2.1.

About this task

Ensure that the minimum RAM and disk space requirements for Unisphere for PowerMax are met. For more information about server requirements, see Pre-installation considerations.

For Linux upgrades to Solutions Enabler 9.2.x, you must perform the upgrade only as the root user (that is, the user running the Solutions Enabler daemons). This is because the 9.1 versions of Solutions Enabler was installed by the root user. Attempting to perform the Solutions Enabler upgrade as a non-root user fails.

Unisphere for PowerMax upgrades are supported from root to root, or non-root to non-root only. The Unisphere for PowerMax Java process run using the same user account that runs the Solutions Enabler daemons.

Before commencing any upgrade process, ensure that backups of all data have been made.

As a safeguard, follow these steps:

Steps

1. Back up the system database and for each registered storage system, back up the performance database.
2. Take note of the systems that are registered for collecting performance data. Record the settings for data collection and diagnostic interval for each one.
3. Export the Performance settings (metric and alert definitions and user templates). This operation requires setting a password to use when you import the saved settings to the new system.

Running the Unisphere for PowerMax upgrade

About this task

This section describes how to upgrade the Unisphere for PowerMax software.

Upgrading Unisphere for PowerMax on Windows

About this task

To upgrade Unisphere for PowerMax using the wizard on Windows:

Steps

1. Access the software, as described in Step 1: Accessing the software.
2. Navigate to the location of the kit and run the appropriate executable file for your operating system.
3. In the **Unisphere for PowerMax Instance Detected** dialog box, click **Yes** to continue and upgrade.
4. A database backup warning displays. Click **OK** to continue with the upgrade. To preserve performance data, ensure that a backup has been performed.
5. In the **Introduction - Upgrade Unisphere for PowerMax** page, click **Next**.
6. In the **Pre-Installation Summary** page, click **Install**.
7. When the upgrade process completes, click **Done**.

Upgrading Unisphere for PowerMax on Linux

About this task

To upgrade Unisphere for PowerMax using the command line on Linux:

Steps

1. Stop the SMAS service, as described in Starting and stopping the SMAS service.
2. Access the software, as described in Step 1: Accessing the software.
3. Navigate to the location of the kit and run the appropriate executable file for your operating system.
4. When prompted to do so, confirm that you want to upgrade the installed features.
5. The installation script displays the current Unisphere version and the version to which you will upgrade. Press **Enter**.
6. A database backup warning displays. Click **OK** to continue with the upgrade. To preserve performance data, ensure that a backup has been performed.
7. In the **Introduction - Upgrade Unisphere for PowerMax** page, click **Next**.
8. In the **Pre-Installation Summary** page, click **Install**.
9. Press Enter through each option in the upgrade process.
   The upgrade process completes.

Next steps

Complete the tasks described in After upgrading Unisphere for PowerMax.

Upgrading Unisphere for PowerMax using silent install

About this task

Unisphere for PowerMax supports silent install upgrades. During an upgrade, a response file and –f options are not required as the installer takes the parameters from previously deployed persistent files. Complete the following steps to perform a silent install upgrade:

Steps

Run one of the following commands:

- For Linux: `./UNIVMAX_version_LINUX_X86_64.bin -i silent`
- For Windows: `UNIVMAX_version_WINDOWS_X86_64.exe -i silent`

NOTE: For Windows, you can use `start /wait` to end the command execution: `start /wait UNIVMAX_version_WINDOWS_X86_64.exe -i silent`

After upgrading Unisphere for PowerMax

About this task

This section describes the tasks to perform after the installation has been completed.

Renewing link and launch client registrations

Steps

If you were using link and launch client registrations to launch Unisphere from ProSphere, or vSphere VSI clients, re-register your host with Unisphere for PowerMax. For more information, see the Unisphere for PowerMax Online Help.

Restoring and verifying system and performance data

About this task

To restore and verify system and performance data, complete the following steps:

Steps

1. Re-register the systems that were collecting performance data before the upgrade.
2. Verify your data collection and diagnostic intervals against the values that are recorded before the upgrade.
3. Restore your performance database backups using the Unisphere for PowerMax console and verify each database.
4. Verify the Performance settings (metric and alert definitions and user templates). If you want to import the Performance settings, use the password that you set when you exported the settings.

   The Unisphere for PowerMax Online Help contains instructions on performing these tasks.

Upgrading Database Storage Analyzer Listener

About this task

To use Database Storage Analyzer packaged with Unisphere for PowerMax 9.2.1, you must upgrade to the 9.2.1 version of the Dell EMC DSA Listener. You can upgrade from version 8.3, or higher.

Install the DSA Listener on a Microsoft SQL Server host. The following Windows Server versions are supported:

- Windows Server 2012
- Windows Server 2012 R2

- Windows Server 2016
- Windows Server 2019

  If using Windows 2019, you must install the OpenSSH client and server separately. Use the instructions provided by Microsoft, at https://docs.microsoft.com/en-us/windows-server/administration/openssh/openssh_install_firstuse, to install and configure OpenSSH. By default, it is installed on port 22, and this is the port that you should use when installing DSA.

You must have Administrator privileges to complete the installation.

Steps

1. In Unisphere for PowerMax, select **Database** > **Administration**.
2. Click **Register**.
3. Select **MS SQL Server**.
4. To download the DSA Listener installer, click the **DSA Listener** download icon.
   The `UNIPMAX_DSA_LISTENER_WINDOWS_X86_64.exe` file downloads in your browser.
5. Copy the downloaded file to the Microsoft SQL Server database host.
6. On the SQL server database host, double-click the executable file to start the upgrade.
7. Proceed through the installation wizard, accepting the default values.
   The installation wizard installs and starts the required services.

4

Uninstalling Unisphere for PowerMax

Topics:

- Before you begin
- Uninstalling from Windows
- Uninstalling from Linux

Before you begin

Before uninstalling Unisphere for PowerMax, note the following:

- Uninstalling Unisphere removes the performance database. The Unisphere online help provides instructions on backing up/restoring the performance database should you want to reuse it later.
- During an uninstallation, the installation program prompts you to export your customized performance settings (metrics and threshold settings only) to a file that can be imported to another Unisphere environment. The Unisphere online help provides instructions on exporting/importing your customized performance settings.
- Uninstalling Unisphere terminates all client sessions to the SMAS server and any SYMAPI operations in progress.

NOTE: Before you uninstall Unisphere for PowerMax, Dell EMC recommends that you unregister any storage systems that are registered for performance data.

Uninstalling from Windows

About this task

You can uninstall Unisphere for PowerMax from a Windows host using either the Windows Add/Remove Programs window, the command line, or a wizard.

Using Windows Add/Remove Programs

About this task

The exact steps that you need to follow might vary from the steps that are listed here, depending on the version of Windows you are using.

To uninstall Unisphere for PowerMax from the Windows Add/Remove Programs window:

Steps

1. From the Windows **Start** menu, select **Settings** > **Control Panel** > **Add/Remove Programs**.
2. In the **Add/Remove Programs** dialog, select **EMC Unisphere for PowerMax** and click **Change/Remove**.
3. Complete the remaining steps in the wizard.

Using the Windows command line

About this task

To uninstall Unisphere for PowerMax using the command line:

Steps

1. Change folder location to *install_dir*\SMAS_EMC_SMAS_installation.
2. Type the following to launch the uninstallation console procedure: `Uninstall_EMC_SMAS_Installation.exe -i console`
3. Follow the steps on screen.

   A few files remain after the uninstallation operation. To remove them, delete the SMAS folder.

Using the Windows wizard

About this task

To uninstall Unisphere for PowerMax using the wizard:

Steps

1. Change folder location to *install_dir*\SMAS_EMC_SMAS_installation.
2. Launch the wizard: `Uninstall_EMC_SMAS_Installation.exe`
3. Complete the steps in the wizard.

   A few files remain after the uninstallation operation. To remove them, delete the SMAS folder.

Uninstalling using silent uninstall on Windows

About this task

To uninstall Unisphere for PowerMax in silent uninstall mode:

Steps

1. Change the location to `install_dir\SMAS\_EMC_SMAS_installation`.
2. Type the following to launch the uninstallation: `Start /wait Uninstall_EMC_SMAS_Installation.exe -i silent`.

Uninstalling from Linux

About this task

You can uninstall Unisphere for PowerMax from a Linux host using either the wizard or the command line.

Using the wizard

About this task

Using the wizard to uninstall Unisphere for PowerMax is not supported on virtual machines.

To uninstall Unisphere for PowerMax from a Linux host using the wizard:

Steps

1. Navigate to the following directory: *install_dir*/SMAS/_EMC_SMAS_installation.
2. Type the following to launch the uninstallation wizard: `./Uninstall_EMC_SMAS_Installation -i swing`
3. Complete the steps in the wizard.

   A few files remain after the uninstallation operation. To remove them, delete the SMAS folder.

Using the Linux command line

About this task

To uninstall Unisphere for PowerMax from a Linux host using the command line:

Steps

1. Navigate to the following directory:

   ```
   install_dir/SMAS/_EMC_SMAS_installation
   ```

2. Type the following to launch the uninstallation console procedure:

   ```
   ./Uninstall_EMC_SMAS_Installation
   ```

3. Follow the steps on screen.

   A few files remain after the uninstallation operation. To remove them, delete the SMAS folder.

Uninstalling using silent uninstall on Linux

About this task

To uninstall Unisphere for PowerMax in silent uninstall mode:

Steps

1. Change the location to `install_dir\SMAS\_EMC_SMAS_installation`.
2. Type the following to launch the uninstallation: `./Uninstall_EMC_SMAS_Installation -i silent`.

5

Appendix A - Technical Notes

Topics:

- Technical notes

Technical notes

This appendix provides technical notes for the advanced configuration of this version of Unisphere for PowerMax.

Managing thin pools

It is recommended that thin pools contain DATA volumes of roughly the same size.

SYMAPI server

If you want to change the SYMAPI server that Unisphere is pointing to, uninstall Unisphere, and then reinstall it.

During the install, you supply the new SYMAPI hostname and network port number.

Local replication management

Replication sessions that are managed exclusively by device files cannot be managed by Unisphere for PowerMax.

Advanced configuration settings

Unisphere provides a dialog box for setting storage system preferences, including debug settings. The debug settings are restricted to Unisphere administrators.

Use the debug settings only at the direction of Dell EMC Customer Service.

Configuring idle user timeout

By default, the client browser logs out automatically after 480 minutes (8 hours). To change this value, complete the following steps:

1. Open the following file for edit:

   ```
   install_dir\jboss\domain\deploy\smc.ear\smc.war\WEB-INF\web.xml
   ```

2. Change the value of the `sageIdleUserTimeoutInMinutes` context parameter to any positive integer, and save the file.
3. If it exists, delete the *install_dir*`\jboss\domain\deploy0` folder.
4. Restart Unisphere for PowerMax.

Alerts

By default, most Unisphere status alerts are disabled. *Configuring alert policies* in the Unisphere for PowerMax help includes instructions on enabling them.

NOTE: For additional information about event and alerts, see the *Events and Alerts for PowerMax and VMAX Users Guide*.

Backups

Consider making regular backups of the system and performance databases, scheduling backup tasks where appropriate and possible. For more information about managing backups, see the *Unisphere for PowerMax online help*.

For more information about downloading database backups from Virtual Appliance manager, see the Admin and Housekeeping section in the *vApp Manager for Unisphere for PowerMax online help*.

Protecting against POODLE SSL v3 vulnerability

To protect your system from the Padding Oracle On Downgraded Legacy Encryption vulnerability, ensure that following is set in your browser:

- SSL v3 protocol is disabled.
- TLS 1.2 protocol is enabled.

For instructions, see the documentation for your browser.

Troubleshooting the "database is in maintenance mode" message

If a database restore, backup, or startup operation is in progress and an attempt is made to start a backup, restore, or registration operation, the following message might be displayed:

```
The following Symmetrixs database is in maintenance mode: storage_system_number
```

To resolve this issue, in Unisphere for PowerMax complete the following steps:

1. Click **Performance** > **Settings** > **Databases**.
2. Wait until the **Status** of the database of the appropriate storage system is set to **OK** before attempting to start another database operation.

6

Appendix B - Security Configuration

Topics:

- Replacing the Unisphere for PowerMax and CA server certificates
- Listing certificates
- Deleting certificates

Replacing the Unisphere for PowerMax and CA server certificates

At installation, the installer generates and installs the self-signed server certificate that is used for HTTPS transport-level security. This appendix describes the steps to replace this certificate with the one that a trusted third party issues.

Step 1: Obtaining the keystore password

About this task

You need the keystore password to complete the following procedure. The keystore password is generated during installation and is stored in the following file: *install_dir*/SMAS/jboss/domain/configuration/host.xml.

Open the file and search for `key-alias="tomcat"`. The password is designated by the `keystore-password` attribute.

Generate a new JKS key/trust store file with the server certificate alias "`tomcat`" and keystore password as described above. The key/trust store must contain all CA certificates needed for full certificate trust chain verification.

Step 2: Generating a certificate request

About this task

To generate a certificate request:

Steps

1. a. For Windows, enter the following commands:

```
cd install_dir\SMAS\jboss\standalone\configuration

install_dir\SMAS\jre\bin\keytool -certreq -alias tomcat^
-file tomcatcert.csr -keystore keystore
```

   b. For Linux, enter the following commands:

```
cd install_dir/SMAS/jboss/standalone/configuration

install_dir/SMAS/jre/bin/keytool -certreq -alias tomcat \
-file tomcatcert.csr -keystore keystore
```

2. When prompted to do so, enter the keystore password.

A file named `tomcatcert.csr` is generated. This file is a certificate request. If the Failed to establish chain from reply error message appears, your CA is not in the keystore's default trust chain (the keystore comes with a default set of trusted CAs).

To import a trusted certificate into your keystore:

a. For Windows, enter the following commands:

```
cd install_dir\SMAS\jboss\standalone\configuration
install_dir\SMAS\jre\bin\keytool -importcert -alias root^
-file root_ca.crt  -keystore keystore
```

b. For Linux, enter the following commands:

```
cd install_dir\SMAS\jboss\standalone\configuration
install_dir\SMAS\jre\bin\keytool -importcert -alias root /
-file root_ca.crt -keystore keystore
```

3. Send the generated certificate request file, `tomcatcert.csr`, to your CA for validation. After the request file has been validated, you should receive a signed certificate back from CA.

Step 3: Importing the CA-signed certificate into the keystore

About this task

Starting with HYPERMAX OS Q1 2015 SR, you can import CA-signed certificates using the CLI or scripts.

In the following instructions:

- `signed_certificate_file` is the signed certificate you received from your CA.
- `password` is the password obtained in Step 1: Obtaining the keystore password
- `CA_certificate_alias` is `tomcat`.

Using the CLI to import the CA-signed certificate into the keystore using CLI

About this task

Starting with HYPERMAX OS Q1 2015 SR, you can import CA-signed certificates using the CLI or scripts.

To import the CA-signed certificate into the keystore using the CLI:

Steps

1. If it is running, stop the SMAS server. For instructions, refer to Starting and stopping the SMAS service
   When X.509 certificate-based authentication is chosen at installation time, the SMAS service is not started, by default.
2. a. For Windows, enter the following commands:

```
cd install_dir\SMAS\jboss\standalone\configuration
install_dir\SMAS\jre\bin\keytool -importcert -alias tomcat^
-file signed_certificate_file.crt -keystore keystore
```

   a. For Linux, enter the following commands:

```
cd install_dir/SMAS/jboss/standalone/configuration
install_dir/SMAS/jre/bin/keytool -importcert -alias tomcat \
-file signed_certificate_file.crt -keystore keystore
```

3. When prompted, enter the keystore password.

Using scripts to import the CA-signed certificate into the keystore using scripts

About this task

To import the CA-signed certificate into the keystore using scripts:

Steps

1. If it is running, stop the SMAS server. For instructions, refer to Starting and stopping the SMAS service
   When X.509 certificate-based authentication is chosen at installation time, the SMAS service is not started, by default.
2. a. For Windows, enter the following commands:

```
cd install_dir\SMAS\jboss\bin
service.bat importcacert password CA_certificate_alias^
signed_certificate_file
```

   a. For Linux, enter the following commands:

```
cd install_dir/SMAS/jboss/bin
initcst.sh importcacert password CA_certificate_alias \
signed_certificate_file
```

Step 4: Starting the SMAS server

About this task

Start the SMAS server. For instructions, refer to Starting and stopping the SMAS service.

Listing certificates

Steps

Type the following commands to list certificates:

- On Windows systems:

```
cd install_path\jboss\bin
```

```
service.bat listcerts password
```

- On Linux systems:

```
cd install_path\jboss\bin
```

```
initcst.sh listcerts password
```

In these examples, *password* is the trust store password.

Deleting certificates

Steps

Type the following commands to delete certificates:

- On Windows systems:

```
cd install_path\jboss\bin
```

```
service.bat deletecert password cert_alias
```

- On Linux systems:

```
cd install_path\jboss\bin
```

```
initcst.sh deletecert password cert_alias
```

In these examples, *password* is the trust store password and *cert_alias* is `tomcat`.

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Appendix C - Configuring SMAS to work in z/OS

Topics:

- Configuring SMAS to work in z/OS

Configuring SMAS to work in z/OS

About this task

These instructions apply to systems running Enginuity 5876 or higher. Install Unisphere for PowerMax on an Open Systems server in a client/server mode. For more information about installing Unisphere for PowerMax, see Installing Unisphere for PowerMax. Define the remote host as the IP address of the mainframe.

To configure SMAS to work in a z/OS environment:

Steps

1. Shut down the services as described in Starting and stopping the SMAS service.
2. Do the following on the mainframe side:
   a. Install Solutions Enabler 8.0 or a later release on the mainframe LPAR. Refer to the *Dell EMC Solutions Enabler Installation Guide* for instructions.
   b. Start the API server, API base daemon, GNS daemon, and event daemon on the mainframe LPAR.
3. Do the following on the qualified Open Systems server:
   a. Make the following changes in the file called `daemon_option`, which is located under *`install_dir`*`/symapi/config` on Linux, or *`install_dir`*`\symapi\config` on Windows:

      Enable the DMN_REMOTE_MODE parameter.

      Set the DMN_REMOTE_ADDRESS parameter to the mainframe IP address.

      If daemon STORSRVD on zOS is not listening on the default port (port 2707), set the DMN_REMOTE_PORT.
   b. Start the STP daemon (storstpd).
   c. Open the `ejb-jar.xml` file, which is found in one of the following locations:

      For Linux:

      `install_dir``/SMAS/jboss/domain/deploy/domain-symm.ear/domain-symm-ejb.jar/META-INF`

      For Windows:

      `install_dir``\SMAS\jboss\domain\deploy\domain-symm.ear\domain-symm-ejb.jar\META-INF`
   d. In the `ejb-jar.xml` file, make the following updates:
      - Set the value of the `StatsConnectionType` entry to `SYMAPI_CONN_TYPE_LOCAL`.
      - Set the value of the `StatsNodeName` entry to the IP address of the Unisphere/API server.
      - Set the value of `StatsNetPort` entry, if it was changed when Solutions Enabler was installed on the Unisphere/API server.
      - Uncomment the XML lines. The beginning comment value is `<!--` The ending comment value is `-->` These values should be deleted.

The following extract is from a sample `ejb-jar.xml` file:

```
<env-entry>
<env-entry-name>StatsConnectionType</env-entry-name>
<env-entry-type>java.lang.String</env-entry-type>
<env-entry-value>SYMAPI_CONN_TYPE_LOCAL</env-entry-value>
</env-entry>
<env-entry>
<env-entry-name>NodeName</env-entry-name>
<env-entry-type>java.lang.String</env-entry-type>
<env-entry-value>198.51.100.90</env-entry-value>
</env-entry>
```

4. Delete the `deploy0` directory under the following directory:

   For Linux:

   install_dir`/SMAS/jboss/domain/`

   For Windows:

   install_dir`\SMAS\jboss\domain\`

   NOTE: Ensure that you do not delete the *install_dir*`/SMAS/jboss/domain/deploy` directory, in which you configured the `ejb-jar.xml` file.

5. Restart the services as described in Starting and stopping the SMAS service.
   When the services restart, the `deploy0` directory is re-created automatically with correct configuration.
6. Launch the Unisphere for PowerMax console. For more information, see Launching Unisphere for PowerMax.